MIL-M-45659C(MU)
11 May 1964
SUPERSEDING
MIL-M-45659B(ORD))
29 June 1962

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M110

1. SC0PE

1.1 This specification covers one type of telescope mount which supports Telescope M103 (see 6. 1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45208	Inspection System Requirements
MIL-I-45607	Inspection Equipment, Supply and Maintenance

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F7659765	Mount, Telescope: Ml 10
LSQAP 7659765	List of Supplementary Quality Assurance Provisions for, Mount, Telescope: M110
IEL-7659765	List of Inspection Equipment Numbers for Mount, Telescope: M110

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

PACKAGING DATA SHEET

MIL-P-14232/P765965 Packaging of Mount, Telescope: M110

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with drawings, material Specifications, and general specifications forming a part of this specification.

3.2 Fabrication. - The mount shall be manufactured in accordance with Drawing F7659765 Mount, Telescope: M110" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to assure inherent quality and performance capabilities.

3.3. General specification. - The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workanship

3.4 Environmental.

3.4.1 Operations temperatures. - The mount shall meet the requirements of 3.6.3 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees. and minus 40 degrees Fahrenheit.

3.4.2 Vibration. - The mount shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no loose or missing parts and the mount shall meet all requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

3.5 Performance.

3. 5.1 Orientation. - With the geometric axis of the gimbal tube parallel to the cylindrical 1936 inch mounting surface, the mount shall be positioned at an angle of 45 degrees plus or minus 10 minutes from the horizontal as shown in Figure I. From the above position the leading edge (front) of the lower locking tooth shall be 20 degrees plus or minus 2 degrees to the right of a vertical plane pas sing through the geometric axis of the gimbal tube.

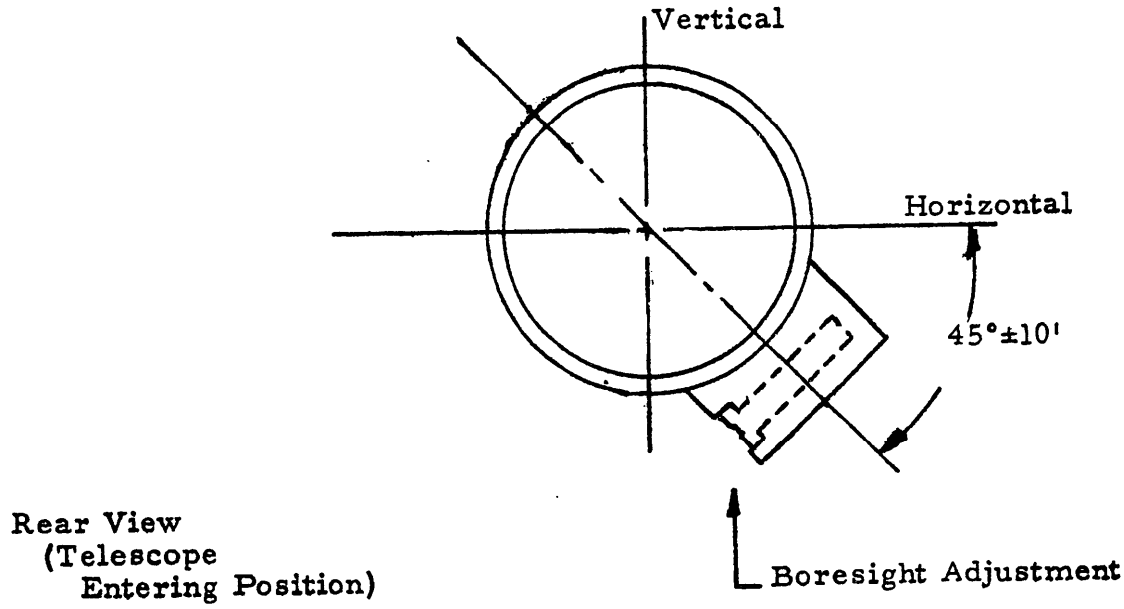


Figure I

3.5.2 Azimuth movement. - Rotation of the azimuth adjustment screw (nearest telescope entering end) shall move the gimbal tube in a horizontal plane through an excursion of a minimum of 13 mils to the right and mils to the left from the midpoint position established in accordance with 3.5.1.

3.5.3 Elevation movement. - Rotation of the elevation adjustment screw (farther from telescope entering end) shall move the gimbal tube in a vertical plane through a minimum excursion of 13 mils elevation and 13 mils depression from the midpoint position established in accordance with 3.5.1.

3.6 Torque.

3.6.1 Ring, externally threaded. - The Ring, 7659771, shall be torqued at final assembly to between 16 and 20 inch pounds with the mount at stadard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

3. 6.2 Adjusting screws. - The torque required for clockwise and counter-clockwise rotation of the azimuth and elevation adjusting screws at standard ambient temperature shall be not Iess than 8 ounce inches.

3.6.3 Azimuth and elevation adjusting screws, - The azimuth and elevation adjusting screws shall be mechanically operable when the mount is exposed to and thermally stabilized at the extreme ambient temperatures of 3.4.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilitv for instruction. - Unless otherwise specified in the contract or Purchase Order the- supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwie specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserve the right to perform any of the inspections set forth in the specifics-tion where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. - Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.2.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in LSQAP 7659765. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic)basis in accordance with Standard MIL-STD-105, and the inspection lever and sampling plans specified in Table I Titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to, be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

TABLE I - CLASSIFICATION OF DEFECT'S

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD =-105

	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.5%		
101. Vibration	3.4.2	4.5.2
102. Orientation	3.5.1	4.5.3
103. Azimuth movement	3.5.2	4.5.4
104. Elevation movement	3.5.3	4.5.5
105. Torque - externally threaded ring (60° to 90°F)	3.6.1	4.5.6.1
106. Torque - azimuth and elevation adjusting screws (60° to 90°F)		
MINOR: NONE		

4. 2.3 Disposition of non-conforming product. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests. - One mount shall be selected at random from each 100 produced, or from each month's production whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Materials	3.1	Per Applicable Drawings and Specifications
Fabrication	3.2	Visual - Per drawings
General Specification requirements	3.3	Visual - Standard Measuring Equipment (SME Specification MIL-F-13926

4. 3.1 Environmental and functional. - Three mounts shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examination and test:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Torque - azimuth and elevation adjusting screws (-40° & +150°F)	3.4.1, 3.6.3	4.5.1, 4.5.6.2.1

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:40Z
Check the source to verify that this is the current version before use.

4.3.2 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-1-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractors delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care maintenance of test equipment shall be in accordance with the "Measuring and Test Equipment" and "Government Furnished Material" requirements of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design. - All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Equipment Numbers IEL 7659765. Athernate designs may be substtituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 Device for conducting performance tests. - A special testing device shall be provided to position the mount in accordance with 3.5 and Figure I. The accuracy of the special test equipment shall be in conformance with 4.4.2.2 and shall be capable of positioning the mount within ten percent of the tolerances specified in 3.5.1 and Figure I. In conjunction with the special test equipment, an auxiliary telescope and a wall target graduated in mils shall be provided to conduct the tests in 4.5.4 and 5.5.4 projector callimator may be utilized in lieu of the wall target for conducting these tests.

6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

4.5 Test methods and procedures.

4.5.1 Operational temperatures. - The mount shall be exposed to the temperatures indicated in 3.4.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of Specification MIL-F-13926. This equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. Temperatures shall be changed gradually to avoid thermal shock. To determine compliance with the requirements of 3.4.1 and 3.6.3 thermal stabilization shall be established at each extreme temperature condition. The mount shall then be subjected to the test procedure of 4.5.6.2.1.

4.5.2 Vibration. - The test shall be performed utilizing a vibrating device capable of meeting conditions of 3.4.2. The accuracy of the applicable test equipment and measuring instruments shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926. The vibrating device shall be equipped with the adapter capable of holding the mount rigidly to the device in it normal operating position. The adapter shall be secured rigidly to the vibrating device by moans of appropriate clamping screws. The mount shall be vibrated in a vertical plane with the motion and frequency applied for the time indicated in 3.4.2. At the conclusion of this vibration, the mount shall be removed from the device, and subjected to a visual and tactile examination for damage such as loose, bent, broken or damaged parts. The mount shall then be subjected to and shall meet all the performance tests specified herein.

4.5.3 Orientation. - The mount shall be oriented on a special testing device as described in 4.4.2.3 which shall be capable of positioning the mount as indicated in 3.5.1 and Figure 1, when properly oriented, the tests of 4.5.4 and 4.5.5 shall be conducted.

4.5.4 Azimuth movement. - The azimuth movement test is accomplished with the aid of a telescope inserted in the mount to check the azimuth movement of the gimbal tube. The mount with the telescope shall be positioned on a special testing device in accordance with 4.5.3. Azimuth movement shall be determined by utilizing a mil graduated wall target. Maxumum horizontal excursion in both directions shall be checked to determine compliance with the requirements of 3.5.2 by movement (rotation) of the azimuth adjustement screw.

4.5.5 Elevation movement. - The elevation movement test is made by the same method employed for making the test in 4.5.4, except this test if for vertical movement of the gimbal tube using the elevation adjustment screw for determining compliance with the requirements of 3.5.3.

4.5.6 Torque.

4.5.6.1 Ring, externally threaded. - The extermally threaded ring 7659771, shall be checked for Compliance with the torque requirements of 3.6, 1 at the standard ambient temperature (60° to 90°F) by means of a commercial torque wrench and appropriate adapt e r.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

4.5.6.2 Azimuth and elevation adjusting screws: The azimuth and elevation adjusting screws shall be checked for compliance with the torque requirements of 3. 6.2 at the standard ambient temperature (60° to 90°F) using the same procedure specified in 4.5.6.1.

4.5.6.2.1 Extreme temperature operation. - The azimuth and elevation adjusting screws shall be checked for compliance with the operability requirements of 3.6.3 at the extreme temperatures specified in 3.4.1. Test equipment and procedures shall be in accordance with 4.5.1. Utilizing a screw driver, the screws shall be manually operable throughout the full range of adjustment while thermally stabilized at each of the temperatures specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14233/P7659765. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The telescope Mount M110 is utilized to mount the direct fire Telescope Ml03 as used in the 90MM Rifle, Recoiless, M67 (MAW).

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing (see 5. 1).
(c) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - Mu

Project No. 1240-A402Y

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T03:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
U.S. ARMY MUNITIONS COMMAND
ATTN: SMUFA-5500
PHILADELPHIA, PA. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M110

This amendment forms a part of Military Specification MIL-M-45659C(MU), dated 11 May 1964.

* Add new paragraph 3.5.4:

"3.5.4 Boresight retention.- A line of sight thru the boresight cross of a simulated telescope shall not deviate by more than 0.25 mils after three engagements of the telescope into the mount. After each engagement this deviation shall be measured relative to the direction of sight prior to the first removal of the telescope.

* Add major Characteristic 107. in Table I as follows:

"107. Boresight retention 3.5.4 4.5.5.1"

Add new paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2 as follows:

"4.1.2 First article (initial Production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 mounts plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.

a record of this inspection, including certificates of Conformance for materials, shall be submitted to the Government for approval. The Govern mant reserves the right to witness the contractor's inspection."

* Paragraph 4.5.3: Delete last sentence and substitute: When properly oriented, the tests of 4.5.4, 4.5.5 and 4.5.5.1 shall be conducted."

* Add new paragraph 4.5.5.1 as follows:

"4.5.5.1 Boresight retention.- The test shall be performed utilizing on alternate test adapter conforming to Drawing F11738151, a viewing telescope conforming to Drawing F5549108 and the drawings depicted thereon.

a. The flat machined surface on the 2 inch diamater of the alinment test adapter shall be positioned in a horizontal plane within 10 seconds, With the viewing telescope inserted into the adapter, the telescope shall be alined so that its optical axis is coincident to the geometric axis of the test adapter. Rotate viewing telescope so that its vertical reticle line is perpendicular to the flat machine surface of the alinement adapeter. The telescope shall then be made secure in- the adapter.

b. The boresight retention test is performed with the mount position and oriented in accordance with 4.5.3. Insert the alinement test adapter, with viewing telescope, into the mount under test. Direct a line of sight through the viewing telescope to the MIL target. Observe the position of the interacting point of telescope's rectcle lines with reference to the intersecting point of the vertical and horizontal target lines. Record the measurement. Any deviation of the line of sight from the inicial recorded measurement shall not exceed the tolerance specified in 3.5.4 when the test adapter, with telescope, is removed and repositioned in the mount three consecutive times."

* The margins of this amendment are marked with an asterisk to indicate where changes from tha provions amendment were made. This was done as a (con- wenience only and the Government aosumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned no evaluate the requirements of this document based on the entire contest irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A279

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:40Z
Check the source to verify that this is the current version before use.